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Knockoff-Guided Compressive Sensing: A Statistical Machine Learning Framework for Support-Assured Signal Recovery

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Abstract

Compressive sensing (CS) is a technique used to recover signals from incomplete or compressed measurements, enabling efficient signal reconstruction. However, commonly used methods like ℓ_1-based approaches (e.g., LASSO) do not directly control the rate of false discoveries when selecting the correct signal components, especially in the presence of correlations between measurements. To address this, we propose **KnockoffCS**, a novel framework that incorporates a knockoff-guided mechanism for support selection. This framework not only controls the false discovery rate (FDR)-a statistical measure to limit false positives-but also employs a separate estimation stage to improve signal recovery. Our theoretical analysis shows that by enforcing FDR control, **KnockoffCS** can reliably identify the correct components and reconstruct signals, even under conditions that are weaker than those typically required by ℓ_1 methods, while maintaining high accuracy. Extensive experiments demonstrate the effectiveness of **KnockoffCS** over existing baselines. In synthetic simulations, **KnockoffCS** significantly improves support recovery (up to 3.9× improvement in F1 score) and reduces reconstruction errors. On real-world datasets, **KnockoffCS** achieves strong downstream predictive performance, surpassing existing CS methods in 71.4% of evaluated modelâ€“dataset combinations, ranking in the top two among CS methods (compared to the compressed baseline, gains are dataset/predictor-dependent). However, performance varies depending on the dataset and predictor, with certain datasets where compressed signal baselines remain competitive. These results highlight **KnockoffCS** as an effective and strong competitive alternative that integrates statistical guarantees into compressive sensing, offering both theoretical reliability and strong empirical results.

Keywords Knockoff · Compressive sensing · Support identification · False discovery rate control

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1 Introduction

Compressive sensing (CS) is a fundamental method showing that sparse signals can be reliably recovered from far fewer measurements than required by the Nyquistâ€“Shannon theorem (Candès et al., 2006). Its key insight is that natural signals often exhibit sparsity, meaning they can be represented using only a small number of non-zero coefficients in an appropriate basis (Candès et al., 2006). The central challenge is thus to recover a high-dimensional sparse signal from limited low-dimensional measurements (Hormati et al., 2009).

A critical aspect of CS recovery, arguably more important than estimating the signal values themselves, is accurately identifying the support set - the locations of the non-zero elements in x. In many applications, such as gene expression analysis, neuroimaging, and radar target detection, determining which components are active is the primary goal (Baron et al., 2008; Kerkouche et al., 2020). For example, in medical imaging, the support set might indicate the locations of abnormalities or regions of interest, while in genomics, it could reveal which genes are relevant to a particular biological process.

Traditional CS approaches tackle support recovery through various strategies. Convex optimization methods, particularly ℓ_1-minimization and its variants, promote sparsity while maintaining computational tractability (Hormati et al., 2009; Tardivel & Bogdan, 2022). Greedy pursuit algorithms like Orthogonal Matching Pursuit (OMP) and CoSaMP iteratively build the support set by selecting elements that maximize correlation with the residual (Wimalajeewa & Varshney, 2016). More recent approaches leverage advances in machine learning, particularly deep neural networks, to learn sophisticated signal structures and recovery mechanisms (Niu, 2022). However, these existing methods suffer from several critical limitations:

- **Lack of False Discovery Control:** While theoretical guarantees exist under restricted isometry property (RIP) or mutual coherence conditions (Baron et al., 2008; Candès et al., 2006), these methods cannot precisely control the proportion of false discoveries in the estimated support set. This is particularly problematic in scientific applications where false positives can lead to misallocation of resources or incorrect conclusions.
- **Sensitivity to Correlation:** Performance often degrades significantly when the measurements or signal components exhibit strong correlation structures (Hormati et al., 2009). Such correlations are common in practice, arising from physical constraints or natural signal properties.
- **Limited Statistical Guarantees:** Most recovery guarantees are worst-case bounds that may be overly pessimistic in practice. Furthermore, they typically lack interpretable statistical significance measures that practitioners can use to assess confidence in their findings.
- **Computational Challenges:** Many methods require careful parameter tuning and can be computationally intensive, particularly for large-scale problems or when high precision is required (Niu, 2022).

Recent advances in statistical machine learning with high-dimensional data have introduced the Knockoff framework as a powerful approach for variable selection (Barber & Candès, 2016). Knockoff matrices are carefully constructed synthetic variables that mimic the

correlation structure of the original variables while being conditionally independent of the response given the original variables. This framework enables principled hypothesis testing for variable selection while controlling the false discovery rate (FDR) - the expected proportion of false positives among all selections (Dai & Barber, 2016). The Knockoff methodology has demonstrated remarkable success across diverse settings, including group-sparse regression (Ren & Barber, 2022), transformational sparsity (Cao et al., 2021), and high-dimensional inference (Machkour et al., 2021). A key strength of the Knockoff framework lies in its model-free approach to FDR control, requiring no restrictive assumptions about noise distributions or signal structures. This flexibility extends to handling complex dependency structures among variables, making it particularly valuable in real-world applications where correlations are inevitable. Furthermore, the methodology scales efficiently to high-dimensional settings while providing interpretable measures of variable importance and selection confidence. These characteristics make Knockoffs an especially attractive tool for modern statistical inference problems where both computational efficiency and statistical rigor are essential.

In this paper, we propose a novel Knockoff-guided compressive sensing framework that bridges the gap between CS support recovery and FDR control, namely **KnockoffCS**. Our approach fundamentally reimagines the CS measurement process through the lens of Knockoffs, enabling precise statistical control over false discoveries while maintaining the computational and sampling efficiency that makes CS attractive. This represents a significant advance over existing methods, providing the first comprehensive solution for statistically principled support recovery in CS. The key innovation of our approach lies in adapting the Knockoff methodology to the unique structure of CS measurements. Unlike traditional Knockoffs designed for regression settings (Barber & Candès, 2016), our construction must account for the compressed nature of the measurements and the specific challenges of the CS inverse problem. We develop a novel measurement matrix knockoff construction that preserves crucial CS properties while enabling FDR control, effectively creating a synthetic control group for hypothesis testing in the compressed domain.

Our primary contributions are:

- **Novel Measurement Knockoffs:** We develop a theoretical framework for constructing measurement matrix knockoffs that preserve the CS measurement structure while enabling FDR control. Our construction carefully balances the competing requirements of Knockoff validity and CS recovery conditions, providing a principled approach to integrating statistical inference into CS.
- **Theoretical Guarantees:** We establish rigorous theoretical guarantees for FDR control in CS support recovery, delineating conditions under which our method maintains the desired false discovery proportion. Our analysis extends existing Knockoff theory (Dai & Barber, 2016) to handle the unique challenges of the compressed measurement setting, including the effects of measurement noise and signal correlation.
- **Efficient Recovery Algorithm:** We introduce a computationally efficient algorithm for joint signal recovery and support identification that leverages both the original and knockoff measurements. Our method achieves superior support recovery accuracy compared to traditional CS approaches while providing guaranteed FDR control, with computational complexity comparable to standard CS.
- **Comprehensive Empirical Validation:** Through extensive numerical experiments on

both synthetic and real-world data, we show that our approach achieves strong downstream predictive performance, surpassing existing CS methods in 71.4% of evaluated modelâ€“dataset combinations, ranking in the top two among CS methods. However, the improvements are not uniform across all datasets and predictors. In particular, we observe significant gains in challenging scenarios involving correlated measurements, structured sparsity patterns, or heavy-tailed noise, especially on datasets like Nanofluid Density, MAGIC Gamma. Nevertheless, in some cases (e.g., Mars dataset for certain predictors), the compressed signal baseline remains competitive, highlighting the need for further tuning depending on the data characteristics.

By equipping compressive sensing with principled FDR control, our framework enables more reliable support recovery and scientific conclusions in practical applications. While methods like SLOPE (Candès & Su, 2015) and T-Rex (Machkour et al., 2024) provide FDR control for sparse estimation in standard regression settings, they are not directly applicable to the CS paradigm. Similarly, existing CS support recovery methods (Niu, 2022) lack formal false discovery guarantees. Our Knockoff-guided framework represents a novel integration of these approaches, providing the first comprehensive method for compressive sensing with guaranteed FDR control on support recovery.

2 Related Work

This section reviews prior work on compressive sensing, false discovery control, and statistical knockoffs, providing the background necessary to situate our framework and highlight the gap it addresses.

2.1 Compressive Sensing and False Discovery Control in Support Recovery

Compressive sensing (CS) enables the recovery of high-dimensional sparse signals from a limited number of linear measurements (Baron et al., 2008; Candès et al., 2006; Candès & Wakin, 2008). Formally, for a sparse signal $\mathrm{x} \in \mathbb{R}^n$ with support $\mathcal{S} = \{i : x_i \neq 0\}$, the observation model is

$$\mathrm{y} = \mathrm{A}\mathrm{x} + \mathrm{w}, \tag{1}$$

where $\mathrm{A} \in \mathbb{R}^{m \times n}$ is the measurement matrix with $m \ll n$, and w denotes noise. The primary objective is to recover x or, more critically, its support $\mathcal{S}$.

Traditional CS recovery methods can be broadly categorized as: *convex optimization*, e.g., ℓ_1-minimization (Basis Pursuit, LASSO) (Hormati et al., 2009; Tibshirani, 1996); *greedy pursuit*, such as Orthogonal Matching Pursuit (OMP) (Wimalajeewa & Varshney, 2016); and *learning-based approaches*, which use deep neural networks to map measurements to estimates (Niu, 2022). While these approaches achieve effective recovery under certain conditions, they generally lack precise statistical error control. In particular, none provides guarantees on the false discovery rate (FDR) in support recovery, often relying instead on heuristic thresholds or tuning rules (Baron et al., 2008; Kerkouche et al., 2020).

Formally, the FDR is defined as

$$\mathrm{FDR} = \mathbb{E}\left(\frac{|\hat{\mathcal{S}} \setminus \mathcal{S}|}{|\hat{\mathcal{S}}|}\right), \tag{2}$$

where $\hat{\mathcal{S}}$ is the estimated support set. Controlling this quantity is essential for reliable inference, especially in scientific applications where false positives are costly.

FDR control has been explored in sparse regression through methods such as Sorted ℓ_1 Penalized Estimation (SLOPE) (Abramovich et al., 2005; Candès & Su, 2015) and T-Rex (Machkour et al., 2024), which provide theoretical guarantees on variable selection. However, these methods are designed for regression models and are not directly applicable to the underdetermined structure of CS.

Overall, while the synergy between FDR control and sparse estimation has been studied in high-dimensional inference (Emery & Keich, 2019), its application to compressive sensing remains underexplored. This gap motivates the integration of FDR-controlling frameworks-most notably Knockoffs-into CS, which we develop in the next section.

2.2 Statistical Knockoffs for Variable Selection

The Knockoff framework, introduced by Barber and Candès (2016), provides a principled approach to variable selection with control of the false discovery rate (FDR). The central idea is to construct *knockoff variables* $\tilde{\mathrm{X}}$ that mirror the correlation structure of the original design matrix X while being independent of the response y given X Dai and Barber (2016). These knockoffs act as negative controls: if a variable is truly null, it should be indistinguishable from its knockoff counterpart.

Two main constructions exist. In the *fixed-X* setting, knockoffs are built to satisfy exact moment-matching conditions relative to a fixed design matrix (Barber & Candès, 2016). In the more flexible *model-X* setting, knockoffs are generated by sampling from an estimated distribution of the covariates (Candès et al., 2018). Model-X knockoffs are especially useful in high-dimensional problems, where fixed-X constructions are often infeasible.

Once knockoffs are constructed, the design matrix is augmented with both original and knockoff variables. Feature importance statistics W_j are computed by contrasting the contributions of each variable and its knockoff, and variables are selected using a data-driven threshold that guarantees FDR control at a target level q:

$$\widehat{\mathrm{FDR}} = \frac{\#\{j : W_j \leq -\tau\}}{\#\{j : W_j \geq \tau\} \vee 1}. \tag{3}$$

Over the past few years, the Knockoff filter has been extended to a wide range of settings, including group sparsity (Ren & Barber, 2022), structural sparsity (Cao et al., 2021), and high-dimensional regression (Machkour et al., 2021). These developments highlight the versatility of Knockoff filter as a general tool for statistically rigorous variable selection. A related work by the authors (Zhang et al., 2025) extends the Knockoff framework to variable selection in over-parameterized regression via recursive knockoff construction and anomaly-based testing. In contrast, the present work studies FDR-controlled support recovery in compressive sensing, where knockoffs are built for the measurement operator and selection is followed by signal reconstruction. Consequently, the statistical mechanisms, assumptions, and theoretical guarantees differ substantially from those in Zhang et al. (2025).

3 KnockoffCS: Knockoff-Guided Compressive Sensing Frameworks and Algorithms

We propose **KnockoffCS**, a framework that integrates statistical knockoff filters into compressive sensing to achieve support recovery with guaranteed FDR control, while maintaining computational efficiency.

3.1 Framework Overview

The main steps of **KnockoffCS** are summarized in Algorithm 1. The framework consists of three phases: (1) *Knockoff construction and filter setup* (lines 1–3), where model-X knockoffs of the measurement matrix are generated and knockoff statistics W_j are computed and sorted; (2) *FDR-controlled support recovery* (lines 4–5), where a data-driven threshold is applied to select the support set under the desired FDR level; and (3) *Signal estimation* (lines 6–7), where the sparse signal is reconstructed on the identified support set. Together, these phases form the **KnockoffCS** framework.

Require: Measurements $\mathbf{y}$, measurement matrix $\mathbf{A}$, target FDR level q
Ensure: Reconstructed sparse signal $\hat{\mathbf{x}}$
Generate model-X knockoffs of the measurement matrix
1: $\tilde{\mathbf{A}} \leftarrow$ `GenerateModelXKnockoffs`$(\mathbf{A})$
Identify the support set $\hat{S}$ from $\mathbf{y}$ using $\mathbf{A}$, $\tilde{\mathbf{A}}$, and target FDR q
2: Compute $W_j \leftarrow |[\mathbf{A}^\top \mathbf{y}]_j| - |[\tilde{\mathbf{A}}^\top \mathbf{y}]_j|$ for all j
3: Sort $|W_j|$ in descending order: $|W_{(1)}| \geq \cdots \geq |W_{(n)}|$
4: Set $T \leftarrow \underset{t>0}{\operatorname{argmax}} \left\{ \operatorname{adjust}\left(\frac{|\{j:W_j \leq -t\}|}{|\{j:W_j \geq t\}| \vee 1} \right) \leq q \right\}$ ▷ data-driven threshold
5: $\hat{S} \leftarrow \{j : W_j \geq T\}$
Reconstruct $\hat{\mathbf{x}}$ using $\hat{S}$ and $\mathbf{A}$
6: $\hat{\mathbf{x}}_{\hat{S}} \leftarrow (\mathbf{A}_{\hat{S}}^\top \mathbf{A}_{\hat{S}})^{-1} \mathbf{A}_{\hat{S}}^\top \mathbf{y}$ or $\hat{\mathbf{x}}_{\hat{S}} \leftarrow (\mathbf{A}_{\hat{S}}^\top \mathbf{A}_{\hat{S}} + \lambda \mathbf{I})^{-1} \mathbf{A}_{\hat{S}}^\top \mathbf{y}$
7: $\hat{\mathbf{x}}_{\hat{S}^c} \leftarrow \mathbf{0}$
8: **return** $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}}_{\hat{S}} \cup \hat{\mathbf{x}}_{\hat{S}^c}$

Algorithm 1 `KnockoffCS`: knockoff-guided compressive sensing

3.2 Knockoff Construction for Compressive Sensing

In compressive sensing, the measurement matrix A is fixed, which would suggest using fixed-X knockoffs (Barber & Candès, 2016). However, such constructions are rarely feasible in high-dimensional regimes. Instead, **KnockoffCS** adopts the model-X knockoff framework (Candès et al., 2018), which generates valid knockoffs by sampling from an estimated distribution of the features. This choice ensures scalability and allows us to rely on established methods, including Gaussian knockoffs (Candès et al., 2018), semidefinite programming relaxations (Barber & Candès, 2016), and deep generative knockoffs (Romano et al., 2020). In Algorithm 1, this step is represented abstractly by the function `GenerateModelXKnockoffs`.

3.3 FDR-Controlled Support Recovery

This phase corresponds to lines 4–5 of Algorithm 1. With the knockoff measurements in place, we construct feature statistics for support recovery. For each index $j \in \{1, \ldots, n\}$, we define

$$W_j = \left|[\mathrm{A}^\top \mathrm{y}]_j\right| - \left|[\tilde{\mathrm{A}}^\top \mathrm{y}]_j\right|, \tag{4}$$

where larger W_j indicates stronger evidence that index j belongs to the true support.

The estimated support set is then defined as

$$\hat{S} = \{j : W_j \geq T\} \quad \text{s.t.} \quad T = \min\left\{t > 0 : \frac{|\{j : W_j \leq -t\}|}{|\{j : W_j \geq t\}| \vee 1} \leq q\right\}. \tag{5}$$

This guarantees that the expected false discovery rate does not exceed the target level q. Here, $|\cdot|$ denotes set cardinality and $\vee$ the maximum operator, with $\vee 1$ ensuring the denominator is nonzero. For an intuitive discussion of this thresholding rule, see the original model-X knockoff paper (Candès et al., 2018).

Other selection rules are also possible, such as the Benjaminiâ€“Hochberg procedure (Benjamini & Hochberg, 1995) or simply choosing the top-ranked variables by W_j. In Algorithm 1, we illustrate the thresholding-based strategy.

3.4 Signal Estimation

This phase corresponds to lines 6–7 of Algorithm 1 and completes Phase 3 of the framework. After identifying the support $\hat{S}$, we estimate the signal coefficients on $\hat{S}$ by solving a regularized least-squares problem:

$$\hat{\mathrm{x}}_{\hat{S}} = \left(\mathrm{A}_{\hat{S}}^\top \mathrm{A}_{\hat{S}} + \lambda \mathrm{I}\right)^{-1} \mathrm{A}_{\hat{S}}^\top \mathrm{y}, \tag{6}$$

where $\lambda \geq 0$: $\lambda = 0$ recovers the standard least-squares estimator, while $\lambda > 0$ corresponds to Ridge regularization for improved numerical stability.

For indices outside the support, we set $\hat{\mathrm{x}}_{\hat{S}^c} = 0$ and combine the two parts to obtain the final estimate:

$$\hat{\mathrm{x}} = \hat{\mathrm{x}}_{\hat{S}} \cup \hat{\mathrm{x}}_{\hat{S}^c}. \tag{7}$$

This construction yields a valid sparse vector in $\mathbb{R}^n$ and can be efficiently solved using methods such as conjugate gradients for large-scale problems.

3.5 Discussions

Remark 1 (Practical Considerations) To apply **KnockoffCS** in practice, several points are worth noting:

- **Parameter Tuning** â€“ The FDR level q should be set according to application needs, balancing discoveries and false positives. The knockoff construction involves a strength parameter s_{ko}, which controls the coupling between each column of A and its knockoff counterpart while ensuring feasibility of the knockoff Gram/covariance constraints (Barber & Candès, 2016; Candès et al., 2018). In our implementation, we do not manually tune s_{ko}; instead, it is automatically selected by the knockoff construction routine using a standard feasible choice. The regularization parameter λ stabilizes matrix inversion without introducing undue bias.
- **Computational Complexity** â€“ Support recovery has complexity $O(mn + n\log n)$ and signal estimation $O(ms^2)$, comparable to standard LASSO. The $O(mn)$ term comes from computing $\mathrm{A}^\top \mathrm{y}$ and $\tilde{\mathrm{A}}^\top \mathrm{y}$, and $O(n\log n)$ from sorting W_j.
- **Extensions** â€“ The framework extends naturally: group knockoffs handle group sparsity, structured knockoffs adapt to known patterns (e.g., tree or spatial), and sequential knockoffs address dynamic data for online FDR-controlled recovery.

4 Main Results

This section develops the theoretical foundations of **KnockoffCS**, formally establishing FDR-controlled support recovery and high-probability reconstruction guarantees.

Consider the compressive sensing model in Eq. 1, where $\mathrm{x} \in \mathbb{R}^n$ is the sparse signal, $\mathrm{y} \in \mathbb{R}^m$ the observation, and $\mathrm{w} \in \mathbb{R}^m$ the noise. We work under the following settings (see Assumption 1):

1. **Signal Properties:** The signal x is sparse, with $\|\mathrm{x}\|_0 \leq s$ and $s \leq m$.
2. **Noise Assumptions:** The noise vector w is Gaussian, $\mathrm{w} \sim \mathcal{N}(0, \frac{\sigma^2}{m}\mathrm{I}_m)$, where $\sigma/\sqrt{m}$ is the noise standard deviation.
3. **Measurement Matrix:** The measurement matrix $\mathrm{A} \in \mathbb{R}^{m\times n}$ and its knockoff $\tilde{\mathrm{A}}$ satisfy a separation condition between original and knockoff interactions with the signal (formalized in Assumption 1).

We then have the following theoretical results.

Proposition 1 (Recovery Guarantees) Our knockoff-guided procedure, **KnockoffCS**, satisfies the following:

1. **FDR Control in Support Discoveries**: The procedure controls the FDR in support discoveries. Given a fixed target FDR level $q \in (0,1)$, the estimated support set $\hat{S}$ is determined by

$$\hat{S} = \{j : W_j \geq T\}, \tag{8}$$

where W_j is the knockoff statistic for the j-th variable and T is a threshold based on q (as defined in (5)). The FDR is defined and controlled as

$$\mathrm{FDR}(\hat{S}) = \mathbb{E}\left[\frac{|\hat{S} \setminus S|}{|\hat{S}| \vee 1}\right] \leq q, \tag{9}$$

where S is the true support of x and $|\cdot|$ denotes set cardinality. Additionally, the probability of correctly identifying the true support is bounded as

$$\mathbb{P}(S \subseteq \hat{S}) \geq 1 - 4s\exp\left(-\frac{(\delta - T)^2}{8\sigma^2}m\right). \tag{10}$$

2. **Reconstruction Error Bound**: Assume the estimator $\hat{\mathrm{x}}$ is defined as in Sect. 2 with $\lambda \geq 0$, and the measurement matrix A satisfies the restricted eigenvalue and limited cross-correlation conditions, together with $|\hat{S}| \leq m$ (as in Assumption 2). Then, with probability at least $1 - 4s\exp\left(-\frac{(\delta - T)^2}{8\sigma^2}m\right) - \exp\left(-\frac{Cs}{8}\log n\right) - \exp\left(-\frac{C|S_f|}{8}\log n\right)$,

$$\|\hat{\mathrm{x}} - \mathrm{x}\|_2 \leq \frac{\sqrt{C}\sigma\left(\sqrt{s} + \sqrt{|S_f|}\right)}{\left(\kappa_{\min}^2 - \gamma\right)}\sqrt{\frac{\log n}{m}}. \tag{11}$$

here, the target FDR level q mediates the trade-off between the bound and its probability: lowering q tends to reduce $|S_f|$ (tightening the bound) while also affecting the inclusion probability.

3. **Refined Reconstruction Error Bound**: There exists $C > \frac{2}{\log 2}$ such that, for a target FDR level $q = c \cdot \frac{s}{m}$ with $c \in (0,1]$ and any $\eta \in (0,1)$, the estimator $\hat{\mathrm{x}}$ satisfies

$$\|\hat{\mathrm{x}} - \mathrm{x}\|_2 \leq \frac{\sqrt{C}}{\kappa_{\min}^2 - \gamma}\left(1 + \sqrt{\frac{c}{\eta}}\right)\sigma\sqrt{\frac{s\log n}{m}} \tag{12}$$

with probability at least $1 - 4s\exp\left(-\frac{(\delta - T)^2}{8\sigma^2}m\right) - \exp\left(-\frac{Cs}{8}\log n\right) - \exp\left(-\frac{C|S_f|}{8}\log n\right) - \eta$.

4. **Expected Reconstruction Error and FDR Level**: For $q = O(\frac{s}{m})$, the estimator $\hat{\mathrm{x}}$ satisfies

$$\mathbb{E}\left[\|\hat{\mathrm{x}} - \mathrm{x}\|_2\right] \leq O\left(\sigma\sqrt{\frac{s\log n}{m}}\right). \tag{13}$$

Together, these results provide FDR control for support recovery and reconstruction guarantees for **KnockoffCS**.

4.1 Support Recovery via FDR Control

Our first result establishes theoretical guarantees for **KnockoffCS**, showing that FDR control provides reliable support recovery under the stated conditions. Full proofs are in the supplement.

Assumption 1 *(Knockoff Procedure and Signal Recovery)* Let $\mathrm{A} \in \mathbb{R}^{m\times n}$ be the measurement matrix with knockoff $\tilde{\mathrm{A}}$,$\mathrm{x} \in \mathbb{R}^n$ be s-sparse with $s \leq m$ and support $S = \mathrm{supp}(\mathrm{x})$, and $\mathrm{w} \sim \mathcal{N}(0, \frac{\sigma^2}{m}\mathrm{I}_m)$. Assume:

1. (*Normalization*) For all j, $\|\mathrm{A}_j\|_2 = \|\tilde{\mathrm{A}}_j\|_2 = 1$.
2. (*Knockoff separation on S*) There exists $\delta > 0$ such that for all $j \in S$,

$$\left|[\mathrm{A}^\top \mathrm{A}\mathrm{x}]_j\right| \;\geq\; \left|[\tilde{\mathrm{A}}^\top \mathrm{A}\mathrm{x}]_j\right| + \delta.$$

This separates original from knockoff contributions on the true support and serves as a substitute for stronger RIP-type assumptions commonly used in ℓ_1-based recovery.

Theorem 2 *(Support Recovery via FDR Control) Suppose Assumption 1 holds. For any target FDR level $q \in (0,1)$, **KnockoffCS** ensures*

$$\mathrm{FDR} = \mathbb{E}\left[\frac{|\hat{S} \setminus S|}{|\hat{S}| \vee 1}\right] \leq q, \tag{14}$$

where $S = \mathrm{supp}(\mathrm{x})$ is the true support and $\hat{S}$ the estimated support set defined in (5). Moreover, the probability of including the true support is bounded as

$$\mathbb{P}(S \subseteq \hat{S}) \;\geq\; 1 - 4s\exp\left(-\tfrac{m(\delta - T)^2}{8\sigma^2}\right). \tag{15}$$

Proof sketch The FDR bound (14) follows directly from the general knockoff filter framework (Barber & Candès, 2015). For the inclusion probability (15), we analyze the knockoff statistic W_j for $j \in S$:

$$W_j = |\alpha_j + \gamma_j| - |\beta_j + \tilde{\gamma}_j|,$$

where $\gamma_j = [\mathrm{A}^\top \mathrm{w}]_j$ and $\tilde{\gamma}_j = [\tilde{\mathrm{A}}^\top \mathrm{w}]_j$ are Gaussian with variance σ^2/m. By Assumption 1, $|\alpha_j| \geq |\beta_j| + \delta$, so $W_j \geq \delta - (|\gamma_j| + |\tilde{\gamma}_j|)$. Applying Gaussian tail bounds (only their marginal Gaussian tails are used thus no independence is required) and a union bound over s indices yields inequality (15). □

4.2 FDR Control and Reconstruction Error

This subsection provides recovery guarantees for the reconstruction error of the estimator $\hat{\mathrm{x}}$ (see Sect. 2 for the estimator with $\lambda \geq 0$). We write $\mathrm{A}_{S'}$ for the submatrix of A with columns indexed by $S' \subseteq \{1, \ldots, n\}$. Full proofs are in the supplement.

Assumption 2 *(Measurement Properties and Support Size)* Let $\mathrm{A} \in \mathbb{R}^{m \times n}$, let $\hat{S} \subseteq \{1, \ldots, n\}$ be the estimated support with $|\hat{S}| \leq m$, and set $S = \mathrm{supp(x)}, S_c = \hat{S} \cap S, S_f = \hat{S} \setminus S$.Assume:

1. (*Restricted Eigenvalue*) There exist $\kappa_{\min}, \kappa_{\max} > 0$ such that for all $S' \subseteq \{1, \ldots, n\}$ with $|S'| \leq m$, we have $\sigma_{\min}(\mathrm{A}_{S'}) \geq \kappa_{\min}$ and $\sigma_{\max}(\mathrm{A}_{S'}) \leq \kappa_{\max}$.
2. (*Limited cross-correlation*) $\|\mathrm{A}_{S_c}^{\top} \mathrm{A}_{S_f}\|_2 \leq \gamma$.

We further assume $\kappa_{\min}^2 > \gamma$, so that the denominator $(\kappa_{\min}^2 - \gamma)$ appearing in Theorem 3 is strictly positive.

Remark (Comparison with ℓ 1-based conditions). Assumptions 1–2 should not be viewed as restrictive limitations of the proposed framework. Rather, they constitute one of the main theoretical strengths of our analysis, as they require substantially weaker structural conditions than those commonly adopted in ℓ_1-based compressed sensing.

In contrast to ℓ1-based compressed sensing analyses, which typically rely on global assumptions such as RIP, irrepresentable conditions, or uniform mutual incoherence, Assumptions 1â€“2 require only local distinguishability on the true support and controlled interference between selected and false supports. They do not enforce uniform near-isometry over all sparse subsets and are therefore strictly weaker than standard ℓ1-based conditions for comparable recovery guarantees.

Theorem 3 *(Reconstruction Error Bound for Estimated Signal) Let S and $\hat{S}$ be the true and estimated supports with $|S| = s$. Define$S_c = \hat{S} \cap S, S_f = \hat{S} \setminus S$, and $S_m = S \setminus \hat{S}$. Under Assumptions 1–2, for any target FDR level $q \in (0, 1)$, there exists a constant $C > \frac{2}{\log 2}$ such that the estimator $\hat{\mathrm{x}}$ satisfies, with probability at least*

$$1 - 4s \exp\Big(-\tfrac{m(\delta - T)^2}{8\sigma^2}\Big) - \exp\Big(-\tfrac{Cs}{8} \log n\Big) - \exp\Big(-\tfrac{C|S_f|}{8} \log n\Big),$$

the error bound

$$\|\hat{\mathrm{x}} - \mathrm{x}\|_2 \leq \frac{\sqrt{C}\, \sigma\big(\sqrt{s} + \sqrt{|S_f|}\big)}{\kappa_{\min}^2 - \gamma} \sqrt{\frac{\log n}{m}}, \tag{16}$$

where $\sigma/\sqrt{m}$ is the noise standard deviation.

Proof Sketch Let $h := \hat{\mathrm{x}} - \mathrm{x}$. By the first-order optimality condition of the least-squares estimator on the estimated support $\hat{S}$,$A_{\hat{S}}^{\top}(A_{\hat{S}} \hat{\mathrm{x}}_{\hat{S}} - \mathrm{y}) = 0$, which links h to the noise and the cross-support terms (see the full proof in the supplement).

Using the restricted eigenvalue and limited cross-correlation assumptions, the cross-term is controlled by $\|A_{S_c}^{\top} A_{S_f}\|_2 \leq \gamma$, leading to the inequality

$\|\mathrm{h}_{S_c}\|_2 + \|\mathrm{h}_{S_f}\|_2 \le (\kappa_{\min}^2 - \gamma)^{-1}(\|A_{S_c}^\top \mathrm{w}\|_2 + \|A_{S_f}^\top \mathrm{w}\|_2) + O(\|\mathrm{x}_{S_m}\|_2)$. The concentration of $A_{S_c}^\top \mathrm{w}$ and $A_{S_f}^\top \mathrm{w}$ follows from the Bernstein's inequality summarized in Lemma 1 of the supplement.

By combining the support-recovery and noise-concentration events via a union bound and using Assumption 2 ($\kappa_{\min} > \gamma$) to ensure $(\kappa_{\min}^2 - \gamma) > 0$, we obtain the overall high-probability error bound stated in Theorem 3. □

Theorem 4 *(Refined Reconstruction Error Bound) Under Assumptions 1, 2, there exists a constant* $C > \frac{2}{\log 2}$ *such that, for an FDR control level* $q = c \cdot \frac{s}{m}$ *with* $c \in (0,1]$ *and any* $\eta \in (0,1)$, *the* ***KnockoffCS*** *estimator* $\hat{\mathrm{x}}$ *satisfies*

$$\|\hat{\mathrm{x}} - \mathrm{x}\|_2 \le \frac{\sqrt{C}}{\kappa_{\min}^2 - \gamma}\Big(1 + \sqrt{\frac{c}{\eta}}\Big)\sigma\sqrt{\frac{s\log n}{m}} \tag{17}$$

with probability at least

$$1 - 4s\exp\Big(-\frac{(\delta - T)^2}{8\sigma^2}m\Big) - \exp\Big(-\frac{Cs}{8}\log n\Big) - \exp\Big(-\frac{C|S_f|}{8}\log n\Big) - \eta.$$

Proof Sketch of Theorem 4 Let $E_\eta := \{|S_f|/|\hat{S}| < q/\eta\}$; by Markov's inequality $\Pr(E_\eta) \ge 1 - \eta$, so that on E_η we have $|S_f| \le |\hat{S}|\, q/\eta$ (see the full proof in the supplement). Conditioning on E_η and using $|\hat{S}| \le m$, Theorem 3 yields that, with probability at least

$$1 - 4s\exp^{-\frac{m(\delta-T)^2}{8\sigma^2}} - \exp^{-\frac{Cs}{8}\log n} - \exp^{-\frac{C|S_f|}{8}\log n} - \eta,$$
$$\|\hat{\mathrm{x}} - \mathrm{x}\|_2 \le \frac{C\,\sigma}{\kappa_{\min}^2 - \gamma}\Big(1 + \sqrt{\frac{c}{\eta}}\Big)\sqrt{\frac{s\log n}{m}}.$$

Choosing any fixed $\eta \in (0,1)$ (e.g. $\eta = \frac{1}{2}$) gives the stated $O(\sigma\sqrt{s\log n/m})$ rate. □

4.3 Expected Reconstruction Error and FDR Level

We next bound the expected reconstruction error as a function of the target FDR level. Full proofs are in the supplement.

Theorem 5 *(Expected Reconstruction Error and FDR Level) Under Assumptions 1–2, and using the high-probability bound in Theorem 3, for an FDR level* $q = c\,\frac{s}{m}$ *with* $c \in (0,1]$, *the* ***KnockoffCS*** *estimator satisfies*

$$\mathbb{E}\big[\|\hat{\mathrm{x}} - \mathrm{x}\|_2\big] \le O\left(\sigma\sqrt{\frac{s\log n}{m}}\right). \tag{18}$$

The above guarantee is obtained without requiring RIP-type global incoherence, highlighting that accurate recovery with FDR control can be achieved under milder structural conditions than those typically assumed in $\ell 1$-based compressed sensing.

Proof sketch Starting from the high-probability bound (16) (Theorem 3), take expectations and apply Jensen's inequality to $\sqrt{|S_f|/|\hat{S}|}$, using $|\hat{S}| \leq m$ (A2.3) and $\mathbb{E}[|S_f|/|\hat{S}|] \leq q$; with $q = c\,s/m$ and absorbing the exponentially small tail from the high-probability event, the claim (18) follows. □

5 Experiments

This section evaluates the performance of **KnockoffCS** through both synthetic simulations and real-world applications, examining its accuracy in signal reconstruction and practical utility across domains.

5.1 Simulation Experiments

This subsection presents controlled simulations to evaluate the effectiveness of the proposed **KnockoffCS** framework in support recovery and signal reconstruction.

5.1.1 Experimental Setup

We run controlled simulations of signal recovery from compressed linear measurements, varying sparsity, number of measurements, and noise levels. Our method is benchmarked against two widely used baselines: LASSO-guided and OMP-guided compressive sensing. The evaluation focuses on FDR control, detection power, and reconstruction accuracy.

Synthetic Dataset Generation

Synthetic datasets are generated according to the standard compressive sensing model $\mathrm{y} = \mathrm{Ax} + \mathrm{w}$, where $\mathrm{x} \in \mathbb{R}^n$ is a sparse signal and $\mathrm{A} \in \mathbb{R}^{m \times n}$ is the measurement matrix. We vary the signal dimension $n \in \{500, 1000\}$, number of measurements $m \in \{50, 100, 200\}$, sparsity level $s \in \{5, 10\}$, and signal-to-noise ratio (SNR) $\in \{2, 10, 30, 50\}$ dB. The sparse signal x is constructed with s nonzero entries drawn independently from $\mathcal{N}(0, 1)$, and the remaining entries are set to zero. The signal is then scaled to unit ℓ_2-norm. Gaussian noise $\mathrm{w} \sim \mathcal{N}(0, \sigma^2 \mathrm{I})$ is added to achieve the desired SNR, with noise level computed via:

$$\sigma = \sqrt{\frac{\|\mathrm{Ax}\|_2^2}{10^{\mathrm{SNR}/10}}}.$$

For each configuration, we generate 20 independent trials with newly sampled signals and measurement matrices. A fixed random seed is used to ensure reproducibility.

Signal and Measurement Matrix Construction

The measurement matrix $\mathrm{A} \in \mathbb{R}^{m \times n}$ is generated with a block-diagonal correlation structure. Specifically, the matrix is partitioned into non-overlapping blocks of size 5, with intra-block entries sampled from a multivariate normal distribution with mean zero and pairwise correlation coefficient of 0.6. Each column of A is normalized to unit ℓ_2-norm. This structured design introduces controlled correlation between features, simulating realistic measurement dependencies. The knockoff selection ratio is fixed at 0.1 for all simulations.

Performance Metrics and Implementation Details

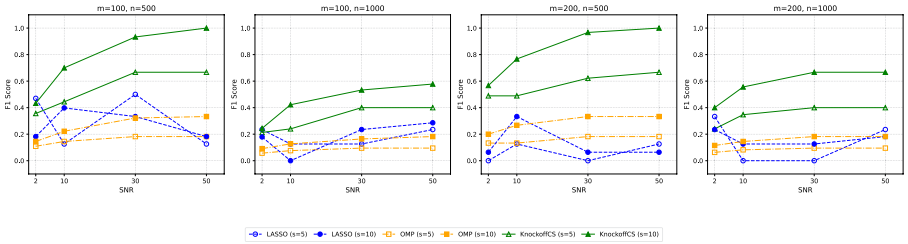


Fig. 1 F1-score comparison across SNR levels for compressive sensing methods

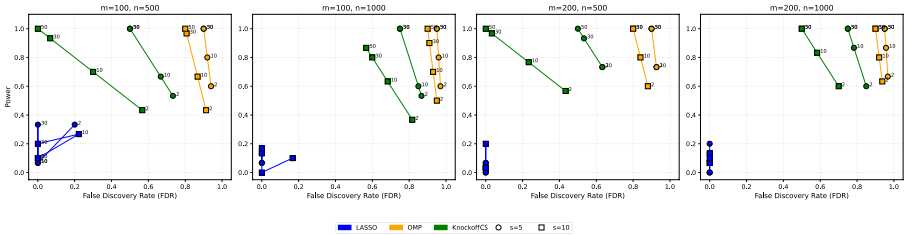


Fig. 2 FDRâ€“power trade-off of compressive sensing methods ($m = 100, n = 500$). Methods closer to the top-left corner indicate better performance, with higher power and lower FDR. Signal-to-noise ratios (SNRs) are annotated near each point (Note: For clarity, overlapping points are labeled only once, and SNR labels are omitted for LASSO in the last three subplots due to frequent overlaps.)

Support recovery is evaluated using three standard metrics: false discovery rate (FDR), defined as the proportion of false positives among selected indices; power (true positive rate), which measures the proportion of correctly identified nonzero entries; and the F1-score, the harmonic mean of precision and recall. Reconstruction performance is assessed via relative error $\|\hat{x} - x\|_2 / \|x\|_2$, and measurement reconstruction error $\|A\hat{x} - y\|_2$.

The Knockoff-based selection in **KnockoffCS** is implemented using the `knockpy` library, adopting a model-X knockoff construction for computational convenience with default parameters; in particular, the strength parameter s_{ko} is automatically chosen by the library. The feature importance statistic is defined as the difference in cross-validated LASSO coefficients between original features and their corresponding knockoffs. The target FDR level q is set to 0.1 throughout all experiments.

Baseline methods (LASSO and OMP) are implemented using `scikit-learn`. For LASSO, we report results using a fixed regularization parameter $\lambda = 0.1$ across all settings.

All simulations are repeated 20 times per configuration to account for statistical variation, and result visualizations in subsequent sections summarize the aggregate trends across these trials. To support reproducibility, the full implementation of the simulation experiments, including data generation, preprocessing, and evaluation code, is publicly available online.[1] All experimental results reported in this paper can be reproduced using the provided code with default configurations.

[1] Anonymous code repository for peer review: https://anonymous.4open.science/r/knockoffcs-9F46/README.md.

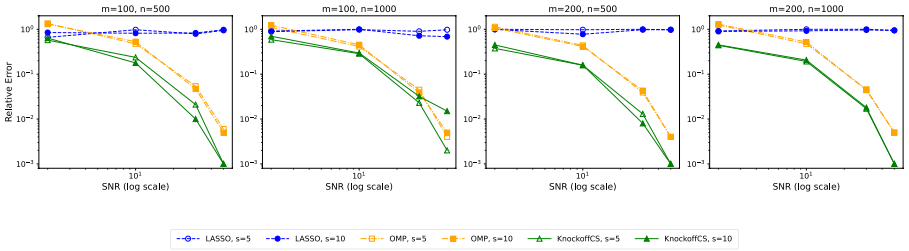


Fig. 3 Relative error comparison across SNR levels for compressive sensing methods

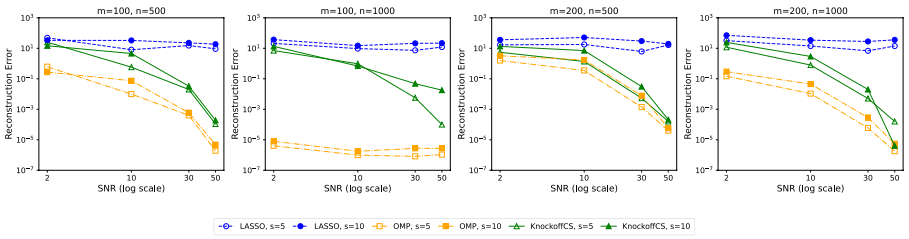


Fig. 4 Reconstruction error comparison across SNR levels for compressive sensing methods

5.1.2 Results of Simulation Experiments

Across noise and sampling regimes, Figs. 1, 2, 3, 4 compare **KnockoffCS** with LASSO and OMP: Figs. 1, 2 report support recovery (F1 and the FDRâ€“power trade-off), while Figs. 3, 4 report reconstruction errors (relative and absolute).

Support Recovery Performance

Our method consistently outperforms baseline approaches in identifying the true support set across all tested configurations. As illustrated in Fig. 1, its average F1-score is approximately 3.9× higher than that of LASSO and 3.45× higher than that of OMP. For instance, under high SNR (50 dB) with $m = 200$ and $n = 1000$, it achieves an F1-score that is 3.66× higher than both baselines. This performance advantage remains stable across different values of m and n, indicating robustness to variations in measurement density and signal dimension.

Figure 2 further demonstrates the superior balance between discovery precision and detection power achieved by our method. In a representative setting with $m = 100$ and $n = 500$, all methods perform poorly at low SNR (2 dB); however, the proposed method exhibits rapid improvement as SNR increases. At 50 dB, it achieves near-perfect power while maintaining an FDR below 0.02. In contrast, OMP exhibits persistently high FDRs, often exceeding 0.8 even at high SNR levels, indicating substantial over-selection and lack of effective support discrimination. Although all methods experience degraded performance under lower sparsity ($s = 5$), our method remains consistently closest to the ideal top-left region in the FDRâ€“power plane.

Signal Reconstruction Quality

We next evaluate the accuracy of the reconstructed signal in both the signal and measurement domains. As shown in Fig. 3, the proposed approach achieves relative reconstruction

errors as low as 10^{-3} under high SNR conditions, significantly outperforming LASSO and OMP, with errors typically ranging between 10^{-2} and 10^{-1}. This reflects a more accurate estimation of the signal magnitude and structure. Figure 4 presents the corresponding measurement reconstruction errors, where our method again demonstrates a clear advantage, particularly in the mid- to high-SNR regimes. These improvements confirm that accurate support identification translates into better downstream signal recovery.

Summary Simulations show that **KnockoffCS** is a robust and accurate CS solution. Its FDR-controlled selection yields precise support recovery and, as a result, superior reconstruction. Compared with greedy and ℓ_1-regularized baselines, which are sensitive to noise and hyperparameter tuning, **KnockoffCS** maintains strong and stable performance across a wide range of configurations, including heavy noise, correlated measurements, and limited sampling.

5.2 Machine Learning with Recovered Signals from Noisy, Low-Dimensional Observations

This section evaluates **KnockoffCS** in a realistic setting: training predictive models on signals recovered from noisy, low-dimensional observations. We start from diverse real-world datasets, inject noise, and apply substantial dimensionality reduction to mimic scenarios with incomplete and corrupted measurements. Our goal is to assess whether signals reconstructed by **KnockoffCS**, leveraging statistically principled support recovery, yield stronger downstream predictive performance compared with signals recovered by established compressive sensing techniques. We then outline the datasets, the construction of noisy compressed observations, the reconstruction procedures, and the evaluation of downstream predictive performance.

5.2.1 Experimental Setup

We construct compressed observations via $\mathrm{y} = \mathrm{Ax} + \mathrm{w}$ with $m \ll d$, add controlled noise, and reconstruct signals using **KnockoffCS** and baselines. Downstream models are trained on the reconstructed signals and evaluated on held-out splits. Implementation choices (noise levels, compression ratios, model training, and evaluation protocols) are specified in the supplement Table 1.

Datasets and Domains We use eight real-world datasets spanning medical imaging, astrophysics, music analysis, materials science, web search, and particle physics. Full statistics and citations (original dimension d_{orig}, added-noise features d_{noise}, compressed dimension m, train/test splits, domain, and task type) appear in Supplementary Table 1. Our comparisons include LASSO, OMP, CLIME (Cai & Liu, 2011), and the Dantzig Selector (Candes & Tao, 2007). All methods are evaluated on identical train/test partitions (typically 80%/20%), as detailed in the supplement.

Observation Synthesis and Signal Reconstruction To simulate the challenging conditions often encountered in practice, we first synthesize a high-dimensional and noisy environment from each real-world dataset. This involves augmenting each original dataset by appending d_{noise} synthetic Gaussian noise features, drawn independently from $\mathcal{N}(0, \sigma^2)$ (where σ is 0.5 times the mean standard deviation of the original features). This step significantly increases the dimensionality of the signal to $n = d_{\text{orig}} + d_{\text{noise}}$ and introduces a

Table 1 Comparison of predictive performance in downstream regression task using signals recovered by different compressive sensing methods with the known measurement matrix (values at the **first places** and the second places among compressive sensing methods)

Predictor Method	CS KnockoffCS	LASSO	OMP	CLIME	Dantzig selector	Compressed	Original
MSE on infrared thermography temperature dataset							
BayesianRidge	2838.257	**2769.503**	3831.489	3173.234	3168.624	2974.272	924.076
LASSO	3008.863	**2985.077**	4251.356	3218.709	3200.671	2991.675	1025.365
Linear	**2837.240**	27915.435	3165.756	108850.513	8850.257	3006.343	925.583
RF	**3089.573**	3816.372	5472.500	4360.481	4214.778	3426.945	925.583
Ridge	2891.314	2889.420	3250.001	2909.050	**2854.285**	3006.340	924.075
SVR	5299.848	**5268.483**	5310.523	5757.232	5774.931	5163.419	4228.817
XGB	**3464.333**	4148.323	6127.383	4420.737	4046.218	3762.099	914.254
MSE on mars asteroid observation dataset ($\times 10^6$)							
BayesianRidge	9.428	**8.579**	9.600	9.265	9.265	9.260	9.162
LASSO	**11.626**	12.007	12.374	11.785	11.785	9.771	9.201
Linear	**11.827**	12.023	12.376	11.786	11.786	9.783	9.413
RF	9.857	**9.060**	10.297	9.473	9.473	9.594	9.477
Ridge	**11.630**	12.023	12.376	11.786	11.786	9.783	9.193
SVR	8.583	**8.538**	9.564	9.292	9.292	9.303	9.148
XGB	**9.934**	10.250	11.502	11.131	11.131	11.395	10.837
MSE on million song dataset							
BayesianRidge	236.896	310.870	**212.671**	220.653	258.239	127.193	112.655
LASSO	**242.865**	325.949	263.020	250.513	253.261	332.422	106.259
Linear	224.801	307.082	729.552	**220.653**	257.837	334.733	106.678
RF	107.418	108.364	**106.758**	147.030	143.723	131.121	94.911
Ridge	192.727	311.854	**174.799**	193.918	246.654	334.733	106.677
SVR	**106.526**	109.067	110.583	162.388	163.020	132.135	141.302
XGB	**118.558**	134.744	125.992	165.279	139.771	153.058	98.159
MSE on nanofluid density prediction dataset							
BayesianRidge	**333.732**	335.056	358.927	431.002	465.867	602.405	43.116
LASSO	328.165	**320.834**	412.115	721.778	452.666	596.149	43.787
Linear	332.696	**325.761**	361.673	739.855	484.902	596.182	43.227
RF	351.855	**292.770**	295.552	468.142	386.882	474.506	5.011
Ridge	328.126	**325.760**	412.706	727.064	484.893	596.182	43.096
SVR	**280.760**	282.845	295.813	440.173	437.418	750.523	431.780
XGB	428.529	361.049	**341.984**	597.856	427.205	530.909	2.124

The **Compressed** column reports performance using raw compressed observations without recovery, while the **Original** column shows performance using the original uncompressed signals

substantial number of irrelevant variables, thereby creating a stringent test for the ability of **KnockoffCS** to accurately identify the true signal support (the original features) while controlling false discoveries among the noise. To prevent any method from exploiting trivial feature ordering and to simulate realistic conditions where informative variable positions are unknown, the augmented high-dimensional signal vector x is randomly permuted.

Subsequently, noisy, low-dimensional observations y are synthesized by applying a common measurement matrix $A \in \mathbb{R}^{m \times n}$ (with entries i.i.d. from $\mathcal{N}(0, 1/m)$) to the permuted high-dimensional signal x, i.e., $y = Ax$. Our experiments focus on the following setting for signal recovery:

- **Recovery with the known measurement matrix** A: For each dataset, we fix a measurement matrix A, ensuring all compressive sensing methods operate under identical conditions when acquiring the low-dimensional observations. During reconstruction, each method attempts to recover an estimate $\hat{x}$ of the (permuted) augmented sparse signal from the observations y using the shared measurement matrix.

For methods reliant on explicit support estimation (OMP and our method), the signal estimate is computed via least squares restricted to the recovered support. A key aspect of the evaluation of **KnockoffCS** is its principled approach to support selection: we select the top 1% of features ranked by their knockoff statistics from the reconstructed $\hat{x}$, which ensures a consistent, interpretable, and stringent sparsity level across datasets.

Prediction Models and Evaluation Metrics The quality of the reconstructed signals $\hat{x}$ from each CS method is ultimately judged by their utility in downstream prediction tasks. To this end, a broad suite of prediction models is employed. For regression, we use Linear Regression, LASSO, Ridge Regression, Bayesian Ridge Regression, Support Vector Regression (SVR), Random Forest (RF), and Extreme Gradient Boosting (XGBoost, XGB). For classification, the models include K-Nearest Neighbors (KNN), Logistic Regression (LOG), Gaussian Naive Bayes (GNB), Support Vector Machine (SVM), Random Forest (RF), Multilayer Perceptron Classifier (MLPC), and XGB. This wide range of models, implemented using `scikit-learn` and `xgboost` with default hyperparameters, ensures that the benefits of **KnockoffCS** are assessed across various learning paradigms. The Knockoff-based selection in **KnockoffCS** is implemented using the `knockpy` library, adopting a model-X knockoff construction for computational convenience with default parameters; in particular, the strength parameter s_{ko} is automatically chosen by the library. Performance is measured using standard metrics: mean squared error (MSE) for regression and accuracy for classification, evaluated on the held-out test set.

To contextualize the effectiveness of recovery methods, we additionally evaluate two non-reconstruction baselines: (1) using the raw compressed observations directly for prediction, and (2) using the original uncompressed signals. This comparative setup enables us to quantify both the predictive loss due to compression and the extent to which different methods can close that gap. The complete implementation is publicly available[2] for reproducibility.

5.2.2 Experiment Results

This subsection presents the empirical results and subsequent analysis based on the aforementioned datasets and experimental settings. The focus is on the downstream predictive performance achieved using signals recovered by **KnockoffCS** and baseline methods.

Results on Regression Tasks In downstream regression tasks (Table 1), signals reconstructed by our method generally facilitated strong predictive outcomes. Across 28 distinct modelâ€“dataset combinations, the proposed method secured a top-two ranking among the compressive sensing methods in 82.1% of instances, highlighting its strong overall consistency compared to LASSO, CLIME, OMP, and Dantzig Selector. For example, on the Infrared Thermography dataset, it achieved top-two performance in six out of seven models.

[2] The code repository is at: https://github.com/xiaochenzhang166/Implementation-of-Knockoff-Guided-Compressive-Sensing.

Table 2 Comparison of predictive performance in downstream classification task using signals recovered by different compressive sensing methods with the known measurement matrix (values at the **first places** and the second places among compressive sensing methods)

Predictor	CS Method	KnockoffCS	LASSO	OMP	CLIME	Dantzig selector	Compressed	Original
Accuracy on HIGGS dataset (2-class classification)								
GNB		**0.506**	0.502	0.498	0.500	0.500	0.546	0.596
KNN		**0.532**	0.522	0.520	0.518	0.518	0.514	0.510
LOG		0.524	0.522	0.516	**0.530**	0.522	0.524	0.644
MLPC		0.536	0.510	0.506	**0.556**	0.532	0.524	0.614
RF		0.526	0.530	0.496	**0.550**	0.522	0.528	0.644
SVM		0.524	0.520	0.514	0.528	**0.544**	0.544	0.630
XGB		0.524	0.520	**0.546**	0.534	0.464	0.572	0.648
Accuracy on kepler exoplanet dataset (3-class classification)								
GNB		**0.396**	0.292	0.316	0.338	0.252	0.236	0.406
KNN		**0.470**	0.392	0.372	0.380	0.446	0.430	0.518
LOG		0.420	0.402	0.404	0.408	**0.428**	0.404	0.758
MLPC		**0.400**	0.366	0.356	0.380	0.390	0.354	0.542
RF		0.398	0.448	0.440	**0.492**	0.482	0.502	0.808
SVM		0.460	0.462	0.458	**0.492**	0.488	0.516	0.468
XGB		0.434	0.402	0.408	0.438	**0.450**	0.474	0.804
Accuracy on MAGIC gamma telescope dataset (2-class classification)								
GNB		0.460	0.565	0.570	0.545	**0.575**	0.640	0.730
KNN		**0.645**	0.625	0.575	0.640	0.605	0.655	0.700
LOG		**0.655**	0.570	0.630	0.605	0.615	0.615	0.745
MLPC		0.580	0.545	0.555	**0.595**	0.545	0.625	0.720
RF		**0.685**	0.680	0.660	0.645	0.670	0.670	0.775
SVM		0.685	**0.695**	0.685	0.660	0.655	0.685	0.755
XGB		**0.655**	0.640	0.605	0.630	**0.655**	0.625	0.810
Accuracy on yahoo! Learning to rank challenge dataset (5-class classification)								
GNB		0.390	**0.470**	0.405	0.420	0.460	0.485	0.425
KNN		**0.430**	0.420	0.265	0.395	0.425	0.510	0.510
LOG		0.460	0.435	**0.485**	0.440	0.465	0.415	0.545
MLPC		0.430	0.430	0.445	0.450	**0.490**	0.460	0.540
RF		**0.530**	0.475	0.475	0.495	0.465	0.550	0.560
SVM		0.500	**0.505**	0.495	0.470	0.495	0.565	0.550
XGB		**0.535**	0.475	0.465	0.510	0.470	0.480	0.575

The **Compressed** column reports performance using raw compressed observations without recovery, while the **Original** column shows performance using the original uncompressed signals

Across the evaluated settings, KnockoffCS achieves lower MSE than using compressed observations directly in 19 out of 28 cases (67.9%). However, these improvements are not uniform; for some datasets (e.g., Mars) the compressed baseline remains competitive or can perform better. Moreover, a gap to the uncompressed benchmark persists in most cases, indicating that reconstruction does not fully recover the original signal quality. On more challenging datasets like Nanofluid Density, it maintained strong performance while avoiding the extreme degradation observed in certain baselines, underscoring its robustness under noisy and ill-conditioned scenarios.

Results on Classification Tasks Our method also demonstrated robust and competitive performance in classification tasks (Table 2). Across 28 classifierâ€“dataset combinations, it

ranked within the top two CS baselines in 64.3% of cases, indicating favorable stability. On the MAGIC Gamma dataset, for instance, the proposed method achieved the highest accuracy among CS methods in over half of the classifiers, outperforming compressed signal baselines and narrowing the gap to the uncompressed reference. On the Yahoo! Learning to Rank dataset, KnockoffCS exhibited mixed performance across different predictors. Specifically, it underperformed the compressed baseline for 5 out of 7 predictors and performed worst for both GNB and MLPC. However, it outperformed the compressed baseline for 2 predictors and ranked among the top methods for 4 predictors, including RF and XGB. These results indicate that while KnockoffCS can outperform compressed observations in some cases, its performance is highly dependent on the choice of predictor. This variability highlights the need for further tuning and adjustments to optimize KnockoffCS in various settings. While performance on other datasets like Kepler and HIGGS varied depending on classifier type, our method showed improvements in predictive quality relative to simple compression alone in 46.4% of the cases, highlighting its potential for better generalization in certain settings. However, there were also instances where KnockoffCS underperformed compared to compressed observations, emphasizing the need for further tuning to optimize its performance across different tasks.

Summary and Observations Across 56 modelâ€“dataset combinations, **KnockoffCS** ranks in the top two among CS methods in 71.4% of cases, indicating strong overall competitiveness relative to the baselines. Compared to the uncompressed originals, reconstructions from KnockoffCS recover a large portion of performance in many settings, although the gap varies and can be large in some cases. The statistically rigorous FDR-controlled selection confers practical value and makes **KnockoffCS** well suited to prediction-driven CS applications. ***Discussion*** A residual predictive gap relative to uncompressed features is expected at low measurement rates under task-agnostic sensing, due to information loss, occasional support errors, and post-selection estimation bias.

This gap can be further reduced by (i) task-aware sensing (Mdrafi & Gurbuz, 2021) that preserves prediction-relevant subspaces, (ii) stronger post-selection decoders (Aeron et al., 2010; Donoho et al., 2011) once support is identified, and (iii) joint sensingâ€“reconstruction optimization under predictive objectives. These extensions are modular and do not alter the FDR-controlled selection mechanism established in this work.

6 Discussion and Conclusions

KnockoffCS is a knockoff-guided compressive sensing strategy that separates support identification from estimation and enforces explicit FDR control in the selection step. This design yields robust reconstruction, especially where conventional approaches are brittle. Theoretically, we show reliable support recovery and reconstruction guarantees under assumptions weaker than those typically required by ℓ_1-based CS. Empirically, simulations demonstrate higher F1 for support recovery and consistently lower reconstruction error than standard baselines across noise, correlation, and measurement regimes. On real datasets, reconstructions produced by **KnockoffCS** yield competitive downstream predictive performance, recovering a substantial portion of the accuracy of uncompressed signals in many settings and ranking among the top two CS methods in 71.4% of evaluated modelâ€“dataset combinations. However, performance is not uniform: the gap to uncompressed signals var-

ies and can be large for some datasets (e.g., Mars). While **KnockoffCS** outperforms established CS baselines in many scenarios, results remain dataset- and predictor-dependent, with noticeable variability across tasks such as Nanofluid Density and MAGIC Gamma. Overall, **KnockoffCS** integrates statistical inference into compressive sensing and offers a principled, practical alternative for prediction-driven CS, combining statistical rigor with strong performance.

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Author Contributions H.X contributed the original idea. X.Z conducted experiments. X.Z and H.X wrote the manuscript. X.Z and H.X share equal technical contribution.

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Data Availability The study uses eight publicly available real-world datasets spanning medical imaging, astrophysics, music analysis, materials science, web search, and particle physics. Each dataset was preprocessed to ensure compatibility with our experimental setup, as detailed in the manuscript. The datasets include: (1) facial and oral temperature data from human volunteers (PhysioNet); (2) Mars Asteroid Observation Dataset (NASA/JPL); (3) Million Song Dataset (ISMIR 2011); (4) hybrid nanofluid density prediction dataset (IEEE Access 2025); (5) HIGGS dataset (UCI Machine Learning Repository); (6) Kepler Exoplanet Dataset (NASA); (7) MAGIC Gamma Telescope dataset (UCI Machine Learning Repository); and (8) Yahoo! Learning to Rank Challenge dataset (PMLR 2011). All datasets are publicly accessible and have been cited appropriately in the manuscript.

Code Availability The code and scripts for reproducing all experiments are available in an anonymized repository for peer review (see footnote) and will be replaced with the official public release upon acceptance.

Declarations

Conflict of interest Not applicable.

Ethical Approval No data have been fabricated or manipulated to support your conclusions. No data, text, or theories by others are presented as if they were our own. Data we used, the data processing and inference phases do not contain any user personal information. This work does not have the potential to be used for policing or the military.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

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